AP PRECALCULUS

Unit 1 • Worksheet 1.8A • Answer key

Worksheet 1.8A • Rational Functions and Zeros

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What are the zeros of $r(x) = \dfrac{x^2 - 9}{x + 2}$?

(A) -2

(B) 3 and -3

(C) 9

(D) 3 only

Solution

Answer: (B). Neither ± 3 is excluded.

Question 2

NO CALCULATOR

What are the zeros of $r(x) = \dfrac{x^2 - 4x}{x - 4}$?

(A) 0 and 4

(B) 4 only

(C) 0 only

(D) none

Solution

Answer: (C). $x = 4$ is excluded.

Question 3

NO CALCULATOR

What is the solution set of $\dfrac{x + 2}{x - 1} < 0$?

(A) $(-\infty, -2)$

(B) $(1, \infty)$

(C) $(-\infty, -2) \cup (1, \infty)$

(D) $(-2, 1)$

Solution

Answer: (D). Test $x = 0$: negative.

Question 4

NO CALCULATOR

What is the solution set of $\dfrac{x-3}{x+1} \geq 0$?

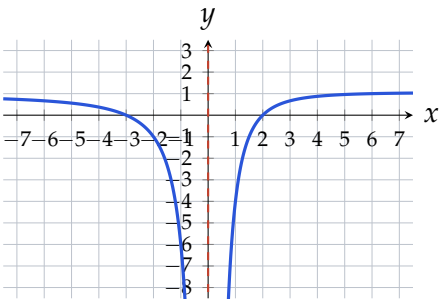
(A) $(-\infty, -1] \cup [3, \infty)$
(B) $[3, \infty)$
(C) $(-1, 3]$
(D) $(-\infty, -1) \cup [3, \infty)$

Solution
Answer: (D). Include $x = 3$, exclude $x = -1$.

Question 5

NO CALCULATOR

The graph of $r(x) = \frac{(x+3)(x-2)}{x^2}$ is shown. Solve $r(x) > 0$.



(A) $(-3, 2)$
(B) $(-3, 0) \cup (0, 2)$
(C) $(-\infty, -3) \cup (2, \infty)$
(D) $(2, \infty)$

Solution
Answer: (C). Above the axis left of -3 and right of 2.

Question 6 (Free response)

NO CALCULATOR

Let $r(x) = \dfrac{x^2 - x - 2}{x^2 - 9}$.

(a) Factor and find the zeros of r.
(b) Make a sign chart and solve $r(x) > 0$.
(c) Solve $r(x) \leq 0$.

Solution
(a) $\frac{(x-2)(x+1)}{(x-3)(x+3)}$; zeros $2, -1$.
(b) Signs on $(-\infty, -3), (-3, -1), (-1, 2), (2, 3), (3, \infty)$: $+, -, +, -, +$. $r > 0$ on $(-\infty, -3) \cup (-1, 2) \cup (3, \infty)$.
(c) $(-3, -1] \cup [2, 3)$.